File 55: Hermes AI SDR Outbound Integration Framework

> ***Venture Scaling & Agentic Integration Guide**: This document establishes the architectural framework for integrating the open-source **Hermes Agent** framework (pioneered by Nous Research) as a self-hosted Sales Development Representative (SDR) for Blue Ridge Stream Restoration & Mitigation LLC. Since Hunter Morris maintains a strict "zero active outreach" brand positioning to preserve his status as an elite stream craftsman, this framework implements a **human-in-the-loop review gate** using Slack/Discord webhook hooks. By routing Hermes-generated geomorphic personalizations through Instantly/Smartlead API channels and syncing opportunities with our open-source Corteza CRM, we create a hyper-scale outbound engine that runs on a private Virtual Private Server (VPS) at minimal cost.*

I. Why Hermes? Self-Hosted Agentic Advantage

Traditional B2B SaaS outreach platforms (such as ZoomInfo or Apollo) charge expensive monthly licensing fees and force developers to send customer data to third-party databases. For a stream restoration venture handling private large-acreage land tracts (\ge 50 acres) and proprietary USACE credit models, data security and local hosting are paramount.
The **Hermes Agent** is an open-source, persistent, self-improving AI agent framework designed to run on a private VPS (Virtual Private Server) or local workstation:

```
                  [HERMES AI SDR ARCHITECTURAL LAYER]

     1. INTELLIGENCE LAYER         2. REVIEW & GATEWAY            3. OUTBOUND DELIVERY
   [Hermes Self-Hosted Agent]    [Slack / Discord Webhook]       [Instantly / Smartlead AP
I]
               |                               |                               |
    Scans registries & HUCs,       Sends drafted proposal to       Pushes greenlit draft to
    diagnoses bank failures,        Hunter's Slack. Reply with      warm inbox for immediat
e
   drafts personalized email       "Yes" or ■ to authorize send     CWA Section 404 delivery
```

- **Persistent Memory**: Hermes remembers landowner interactions, HUC-8 supply gaps, and past email conversion rates across sessions, dynamically adapting its copywriting style.
- **Skill Acquisition**: Hermes automatically refines its own "Skills" locally. For Hunter's business, we package our custom geomorphic soil-loss equations and outfitter-aligned copywriting hooks as reusable Python modules.
- **Zero Data Dilution**: All landowner coordinates, tax records, and contact histories remain protected inside our own self-hosted infrastructure.

II. The 4-Tier Hermes SDR Integration Framework

To operationalize the Hermes Agent framework, we structure the outbound stack into four cohesive layers:

1. The Intelligence Layer (Hermes Core Agent)

The central cognitive engine, running on a local Docker container or private Linux VPS (Ubuntu 22.04 LTS). It runs two specialized "Skills" we have developed:

- **Geomorphic Assessment Skill**: Connects to the local landowner database ([Georgia_B2B_Outreach_&_Referral_Directory.xlsx](file:///Users/irvani/Library/CloudStorage/GoogleDrive-hadi.irvani@gmail.com/My%20Drive/ACR%20Stream%20Restoration/10_Outreach_&_Relationship_Banking/Georgia_B2B_Outreach_&_Referral_Directory.xlsx)) or reads GIS coordinate vectors. It automatically calculates lateral bank erosion loads using our bulk density constant of 85.0 lb/ft^3 and projects stream credits generated under the Savannah District SOP.
- **Angler Sourcing Copywriting Skill**: Generates highly personalized outreach copy. It weaves in Hunter's local outfitting pedigree (managing a team of 12 to 20+ professionals) and contrasts his geomorphic trout craftsmanship against nameless conglomerates that clear-cut streams.

2. The Verification Gate (Human-in-the-Loop Slack/Discord Webhooks)

To protect the high-status authority of Hunter's brand, Hermes is blocked from direct outgoing email delivery. It routes all completed drafts to Hunter's private Slack or Discord channel:

- Hermes posts a structured card:

```
    ■ NEW LANDOWNER PROPOSAL DRAFTED ■
    - Landowner: Roya Irvani (Anderson Creek, 1,500 LF)
    - HUC-8 Basin: Etowah HUC 03150104 (Dawson County)
    - Soil Loss Diagnosed: Severe bank scour (281.25 Tons/Year prevented)
    - Landowner Payout Split: $696,600.00
    - Draft Outreach: "Dear Roya, Hunter Morris noted your beautiful parcel along Anderson
 Creek..."

    ■ Reply 'YES' or react with ■ to authorize immediate push to Instantly.
```

- Hunter Morris simply replies with "Yes" or reacts with a thumbs-up emoji. Hermes detects the webhook trigger, closes the review state, and pushes the draft to the delivery queue.

3. The Deliverability Layer (Smartlead / Instantly API Integration)

Because cold email deliverability requires secondary domains, proxy rotation, and mailbox warm-ups, Hermes delegates sending to specialized platforms via REST APIs:

- Upon Hunter’s Slack approval, Hermes makes a POST request to the **Instantly API** (`https://api.instantly.ai/v1/lead/add`):

```
    {
      "api_key": "instantly_api_sec_key",
      "campaign_id": "anderson_creek_sourcing_campaign",
      "leads": [
        {
          "email": "roya.irvani@example.com",
          "first_name": "Roya",
          "last_name": "Irvani",
          "custom_variables": {
            "waterbody": "Anderson Creek",
            "huc_basin": "Etowah HUC 03150104",
            "sediment_tons": "281.25",
            "payout": "$696,600.00",
            "custom_body": "Hunter Morris noted your beautiful parcel along Anderson Creek
..."
```

```
            }
        }
    ]
}
```

- The email is sent from our secondary warming domain (`@restorerivers.com` or `@blueridgestreams.com`) to protect our primary domain.

4. The Opportunity Sync Layer (Corteza CRM & Calendar Automation)

When a landowner replies:

- Hermes continuously monitors the mailbox webhook. It processes incoming reply sentiments using natural language classifiers.
- **Positive Reply**: If the landowner replies ("Sure, let's talk next Tuesday"), Hermes:

1. Creates a new "Qualified Lead" inside our open-source **Corteza CRM** (as audited in File 39).
2. Pushes a calendar event invite for a "Free Geomorphic Field Walk" to Hunter's Google Calendar.
3. Pre-drafts a Georgia UETA-compliant Joint Venture Sourcing Agreement template for Hunter's review.

III. Step-by-Step Self-Hosted Deployment Guide

Hunter Morris and his graduate interns can deploy this Hermes AI SDR framework locally in three phases:

Phase 1: Initialize the Docker Container

Deploy a Linux VPS (such as a 4-vCPU Contabo instance for $8.40/month) running Docker and pull the Nous Research Hermes Agent image:

```
# Update local packages
sudo apt update && sudo apt upgrade -y

# Install Docker
curl -fsSL https://get.docker.com -o get-docker.sh
sh get-docker.sh

# Clone Hermes Agent repository & spin up container
git clone https://github.com/NousResearch/Hermes-Agent.git
cd Hermes-Agent
docker compose up -d
```

Phase 2: Register Local Custom Skills

Mount our custom geomorphic and copywriting tools as Python plugins. Create `/skills/blue_ridge_skills.py` inside the container:

```
# Custom Fluvial calculations mounted to Hermes runtime
def get_geomorphic_yield_metrics(length: float, erosion_rate: float, height: float) -> dic
t:
    bulk_density = 85.0
    sediment = (length * height * erosion_rate * bulk_density) / 2000.0
    credits = int(length * (4.3 if erosion_rate >= 1.0 else 3.8))
    value = credits * 110.0
    return {
        "sediment_prevented": round(sediment, 2),
```

```
        "credits": credits,
        "total_value": round(value, 2),
        "payout_split": round(value * 0.30, 2)
    }
```

Phase 3: Setup Slack Event Subscriptions

1. Go to api.slack.com and create a new Slack app: "Blue Ridge SDR Bot".
2. Enable **Incoming Webhooks** and paste the webhook URL into Hermes config: `SLACK_WEBHOOK_URL="https://hooks.slack.com/services/..."`.
3. Under **Event Subscriptions**, enable message event listens and point the Request URL to your Hermes VPS server IP: `https://your-vps-ip/api/v1/slack/events`.
4. Now, any response to the Slack bot will trigger immediate action.

IV. Core Performance KPIs for Hunter's EOS Scorecard

To monitor the efficiency of the Hermes SDR bot on Hunter's weekly Traction EOS Scorecard (as outlined in File 52), track these 4 metrics:

Metric	Target	Description	Tool Source
Weekly Sourced Leads	**15 Reaches**	Unique landowners identified and analyzed.	Hermes Database Scan
Slack Approval Rate	**100%**	Drafts greenlit by Hunter Morris.	Slack Webhook Logs
Email Open Rate	**> 65.0%**	Delivered emails opened by landowners.	Instantly Analytics
Field Audits Booked	**2 Audits/Wk**	Confirmed face-to-face creekside site walks.	Google Calendar Sync

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Developed by Blue Ridge Stream Restoration Technical Sourcing Operations. Pushed to remote origin under main.